David Andrews

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Education

Georgia Institute of Technology, Atlanta, GA: *B.S. in Computer Science* Aug. 2023 – May 2026
GPA: 4.0/4.0
Courses: DL, ML, Systems for ML, Intro to AI, NLP, Perception and Robotics, Algorithms, OS

Experience

Georgia Tech, Chao Zhang Lab: *Lead Undergraduate Researcher, Artificial Intelligence* Aug. 2024 – Present

- Leading a team of 3 building a general tool-using agent framework with Docker sandboxes and an MCP interface (browser, terminal, code execution), supporting multi-turn rollouts across math, search, SWE-bench, and puzzle tasks.
- Trained multi-turn LLM agents with GRPO on SWE-bench, SWE-Smith, and SWE-Gym, adding reward-hacking defenses, tool-use checks, and rollout instrumentation to improve training stability.
- Scaled asynchronous PPO/GRPO/RLVR training using AReaL on Ray to maximize throughput under high tool latency, coordinating rollout workers with sandboxes on a Kubernetes cluster across 8-GPU machines.

Georgia Tech, Qirun Zhang Lab: *Undergraduate Researcher, Systems* Jan. 2026 – Present

- Leading development of an LLM agent harness for automated bug finding and fixing in LLVM, with specialized tool skills for navigating large-scale systems codebases with long build times.
- Investigating agent-based commit bisection to accelerate identification of bug-introducing changes, reducing manual debugging effort in complex compiler histories.

Georgia Tech, EIC Lab: *Undergraduate Researcher, Hardware Design* Feb. 2026 – Present

- Curating a ∼250-task SWE-bench-style benchmark for Verilog/RTL hardware design derived from real-world GitHub PRs, using Verilator, Yosys, and Icarus Verilog for functional verification.
- Developing an agent harness to evaluate frontier models on industry-relevant hardware design tasks, targeting state-of-the-art performance on the benchmark.

DuckAI: *AI Research Lead* Aug. 2023 – Present

- Engineered **DuckTrack**, a cross-platform computer-use capture tool adopted by OSWorld and OpenCUA for benchmarking and foundation model training.
- Led development of **PRMBench**'s full-stack annotation platform (React, FastAPI, SQLite), managing a team of annotators and improving labeling speed and inter-annotator agreement for process reward model datasets.
- Built large-scale TTS data pipelines (YouTube and podcast ingestion, upscaling, VAD, diarization) and contributed optimizations to insanely-fast-whisper to accelerate transcription for speech datasets.

Selected Projects

DuckTrack Nov. 2023 – Jan. 2024

- Extended and maintained precise cross-platform capture of screen, keyboard, and mouse events, synchronizing native input APIs with OBS to achieve millisecond-level sub-frame alignment for computer-use datasets.

Emergent Misalignment Oct. 2025 – Jan. 2026

- Replicated and investigated emergent misalignment arising from benign fine-tuning of LLMs; findings published on LessWrong.

AlphaEvolve Extensions May 2025 – Sept. 2025

- Extended OpenEvolve, an open-source replication of Google DeepMind's AlphaEvolve, with new combinatorics domains including circle-packing and sphere-packing; built a parallelized parameter sweep system using Optuna.
- Co-contributed to an upstreamed PR adding MLIR attention kernel optimization tasks, achieving a 1.32x+ kernel speedup via evolved transformation parameters.

Skills

Languages: Python, Rust, C, C++, Go, JavaScript, Bash, Verilog
ML: PyTorch, vLLM, SGLang, Ray, AReaL, veRL, HuggingFace
Systems: Linux, Docker, Kubernetes, SLURM, AWS, FastAPI, SkyPilot, uv, Git